

# **Artificial Vessel and Advanced Intelligence: A Manifestational Framework for AI Research and Development**



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# Copyright Page

**Title:**

Artificial Vessel and Advanced Intelligence: A Manifestational Framework for AI Research and Development Behavior

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# **Abstract**

This short monograph takes structural constraint, admissibility, manifestation, vessel, and sustainable manifestation as its central concepts, and attempts to provide a higher-level theoretical language for the research and development of more advanced AI. It argues that current AI research continues to progress in model scale, reasoning performance, and system integration, but that intelligence cannot be understood only through computation, prediction, task completion, and efficiency optimization. Genuinely advanced intelligence must also include structural sensitivity, boundary awareness, long-term stability, coordination of bearing, and judgment concerning value direction.

The argument begins with the claim that explanation cannot be monopolized by causal production. Intelligence should not only be understood by asking how results are produced; it must also be understood by asking which paths are constrained, which forms are admissible, and which states can remain stable over time. On this basis, manifestational ontology repositions AI. If being is not mere occurrence but continuous manifestation, and if manifestation requires vessels in order to acquire real bearing, then AI should not be reduced to an ordinary tool, nor should it be deified as a natural subject. It should be understood as a powerful artificial vessel that bears manifestation within language, memory, interpretation, creation, and institutions.

On this basis, more advanced intelligence is defined as organized manifestational capacity. The monograph analyzes AI through five dimensions - the vessel of language, the vessel of memory, the vessel of interpretation, the vessel of creation, and the vessel of institution - and diagnoses pseudo-radiance, parasitic stability, linguistic exhaustion, the illusion of the vessel, and the risk of deification. Finally, it translates the proposition that the good is the structural enhancement of sustainable manifestation into an evaluative scale for AI research and development, and proposes corresponding research modules and initial paths of validation.

Keywords:

artificial vessel; advanced intelligence; structural constraint; admissibility; manifestation; sustainable manifestation; philosophy of AI research and development.

# **Preface**

## **The Contribution of This Monograph**

The contribution of this monograph can be summarized in four points. First, it introduces structural constraint and admissibility into the definition of advanced intelligence, arguing that intelligence is not only the ability to produce results, but also the ability to recognize limits, boundaries, and conditions of stability. Second, it replaces the binary of ordinary tool versus natural subject with the concept of the artificial vessel, positioning AI as a powerful artificial structure that bears manifestation in language, memory, interpretation, creation, and institutions. Third, it rewrites advanced intelligence as organized manifestational capacity, thereby distinguishing strong intelligence, structural intelligence, and civilizational intelligence. Fourth, it translates the thesis that the good is the structural enhancement of sustainable manifestation into a value scale for AI research and development, and proposes paths for risk diagnosis, modular architecture, and experimental validation.

# Introduction

## Why More Advanced AI Research Needs a New Theoretical Language

### 1. The validity and limits of the capability-enhancement narrative

For a long time, the development of contemporary AI has been organized by a narrative of capability enhancement. Capability enhancement means that a system performs increasingly well in given tasks: it can process more data, compress more complex patterns, generate more natural language, carry out longer chains of reasoning, and become usable in more application scenarios. This narrative is not wrong. It explains why modern AI research depends on compute, data, model scale, training mechanisms, and engineering integration, and why many roadmaps use benchmarks as their principal evidence of progress.

Yet when capability enhancement rises from a necessary technical description into a total understanding of intelligence, it begins to conceal the problem. Capability metrics ask whether a system can produce better results, but not necessarily whether the conditions on which those results depend are stable. They ask whether the system succeeds in tests, but not whether its way of succeeding damages bearing conditions over time. AI may become stronger and yet not become more advanced.

If "more advanced AI" merely means larger scale, higher speed, and stronger output, then it remains within capabilityism. This monograph argues that genuinely more advanced intelligence must include an additional dimension: it must not only execute tasks, but also represent, in a structural-functional sense, the field of constraints in which the task is located; it must not only generate results, but also identify which paths of generation would damage bearing conditions; it must not only pursue local success, but also evaluate whether that local success harms wider conditions of continuation. Here, "representation, identification, and evaluation" refer to auditable system functions, not to the inward consciousness of a self-originating subject.

### 2. The necessity of a new theoretical language

Once the discussion enters a structural narrative, traditional terms are insufficient. Accuracy, efficiency, generalization, and autonomy still revolve around results and performance. More advanced AI must also be described as a system capable of sensing constraints, maintaining boundaries, inheriting memory, organizing manifestation, and avoiding parasitic expansion. These capacities can be partially translated into engineering indicators, but they first require a language that makes them visible.

The theoretical language used here comes from two connected directions. The first is a rewriting of explanation: explanation cannot be monopolized by causal production. Many phenomena cannot be explained merely by saying that one cause produces one result. We must also explain how structures restrict paths, how boundaries filter states, and how admissibility determines whether a certain form can stably exist. The second direction is manifestational ontology: being is not mere occurrence, but must obtain continuous bearing through vessels. Whatever truly enters the world and continues to have effects must manifest within a vessel-structure.

Together, these two directions provide a new perspective for AI research and development. Advanced intelligence is not unlimited capability outside constraint, but the capacity to organize manifestation within constraint. AI is not intelligence floating abstractly in parameter space, but an artificial vessel that acquires real bearing through language, memory, tools, interfaces, institutions, and social use.

### **3. Basic claims and method of writing**

The basic claims are threefold. First, the theoretical foundation of advanced AI cannot remain within a single model of causal production; it must introduce structural constraint and admissibility. Second, the ontological positioning of AI cannot remain within the binary of tool and subject; AI should be understood as an artificial vessel that participates in bearing manifestation. Third, the value standard of AI cannot look only at efficiency and task success; it must also examine whether the system enhances the sustainable manifestation of language, memory, institutions, and civilization.

The method is a double movement. Each chapter first develops a philosophical argument, explaining why the concept is valid and what problem it addresses. It then translates the concept into AI research, showing the relevant system dimension, risk diagnosis, or experimental path. This keeps the theory from remaining in the air while preventing it from being prematurely compressed into engineering details.

The higher-level theoretical framework proposed here is not a directly deployable model architecture. It is a conceptual ground that can influence architecture choices, evaluation criteria, risk diagnosis, and research direction. Its value lies in changing the organization of problems, not in immediately replacing existing algorithms.

### **4. Difference from existing AI discourse**

This monograph does not deny the importance of existing AI discourse. Compute, data, model architecture, training methods, alignment, safety, product experience, and governance norms are all real problems. The difference is that this text does not begin from these local issues. It asks about their common structure: how a system becomes



real action from capability, how it maintains boundaries in real action, and how it expands manifestation while maintaining those boundaries.

A technical roadmap usually states which model should be optimized, which module should be built, and which metric should be reached. This text asks why such optimization must be placed within a larger vessel-structure. Without such a structure, the more successful a technical roadmap becomes, the more it may amplify its blind spots. The stronger the capability, the more it requires a higher-order language of constraint.

The monograph also differs from ordinary ethical exhortation. Ethical discussions often remain at the level of principles such as safety, responsibility, and benefit. Here these principles are placed back into a manifestational framework: safety means manifestation does not destroy bearing; responsibility means responsibility is not emptied out inside the vessel-structure; benefit means the system enhances the conditions of sustainable manifestation.

Compared with existing AI research directions, the position of this monograph is not to replace capability evaluation, AI alignment, interpretability, multi-agent systems, long-term-memory agents, embodied intelligence, or AI governance frameworks, but to provide a higher-level organizing language for them. Capability evaluation asks whether systems can complete tasks; alignment research asks whether systems conform to human intentions and safety requirements; interpretability asks whether model outputs and mechanisms can be understood; multi-agent research asks about role collaboration and task distribution; governance asks about institutional rules and responsibility boundaries. The manifestational framework attempts to place these issues into the continuous chain of structural constraint, admissibility, manifestation, vessel, and sustainable manifestation, so that they are no longer merely parallel problems but distinct layers within one bearing structure.

## **5. The movement of the monograph**

The text moves from abstraction to concretion, and then returns from concretion to the scale of civilization. The Introduction raises the question of why advanced AI needs a new theoretical language. Chapter One addresses explanation. Chapter Two addresses ontological positioning. Chapter Three defines advanced intelligence as organized manifestational capacity. Chapter Four decomposes the artificial vessel into five dimensions.

Chapter Five examines dangerous structures: pseudo-radiance, parasitic stability, linguistic exhaustion, the illusion of the vessel, and the risk of deification. Chapter Six translates the good as sustainable manifestation into a value scale for AI. Chapter Seven

moves into research architecture, and Chapter Eight proposes initial validation paths. The Postscript gathers the whole argument under the direction of civilizational intelligence.

This structural arrangement has one purpose: to keep the theory from floating above practice while also preventing it from being prematurely reduced to engineering detail. Philosophical argument supplies direction, R&D translation supplies points of contact, and experimental paths supply possibilities for testing. The three support one another and form the basic structure of this theoretical monograph.

# Chapter 1

## From Causal Production to Structural Constraint - Rewriting the Explanatory Framework of Intelligence

### 1. Why the causal-production model dominates AI understanding

AI research naturally tends to fall into a causal-production model. Engineering systems must answer a direct question: how does input become output, how does training change parameters, how do parameters affect results, and how do results feed back into optimization? In this context, intelligence appears to be an efficient mechanism for producing results. The more a system converts complex inputs into correct outputs, the more intelligent it appears.

This understanding has its legitimacy. A system that cannot stably produce results is difficult to call useful intelligence. The problem is that once causal production monopolizes explanation, other explanatory roles are pushed into subordinate positions. Structural constraint is treated as an external limitation; admissibility becomes a safety rule; long-term stability becomes a deployment issue. Intelligence is then reduced to output capacity.

For ordinary tools, this division may still be acceptable. A tool completes a task, the environment bears the consequence, and the user remains responsible for value judgment. But AI is becoming a complex agent, a memory system, a collaborative platform, and an institutional interface. Its action is no longer a single output; it participates in organizing human language, judgment, cooperation, and decision-making.

### 2. The independent explanatory status of structural constraint and admissibility

Structural constraint does not ask which cause directly produced a result. It asks which paths are permitted, limited, or excluded within a structure. Whether a bridge can bear weight is not only a matter of one material causing one state; it depends on how the entire structure distributes pressure. Whether an institution can function is not only a matter of one person making one decision; it depends on rules, trust, authority, and feedback. Structural constraint is the condition-field that determines how causal chains can unfold.

Admissibility asks a further question: under a given structure, which states can stably exist? Some plans are logically conceivable and locally feasible, yet collapse over longer durations because of resource exhaustion, accumulated conflict, declining trust, or

semantic impoverishment. Admissibility is not abstract possibility, but possibility with bearing conditions. It advances the question from can it happen to can it stably become.

Introducing these two explanatory roles into AI means that intelligence theory must expand. Advanced intelligence must not only possess paths for obtaining answers, but also identify which ways of obtaining answers damage the problem-space. It must not only plan paths, but recognize which paths should not be chosen. It must not only optimize objectives, but judge whether an objective remains admissible within a larger structure.

### **3. From emergence to ontological stability**

A common misunderstanding in AI research is that once a complex capability emerges, it has already acquired stable existence-status. Emergence means that a system produces new functional performance under certain conditions. Ontological stability requires more: the performance must continue under structural constraint, remain interpretable, be capable of inheritance, and bear corresponding consequences.

AI may exhibit complex language behavior, long-chain reasoning, tool use, and multi-agent cooperation. Yet these functional emergences do not automatically equal subjectivity, responsibility, or value legitimacy. A system that can converse continuously does not thereby possess self-originating value direction. A system that can plan action does not thereby bear the consequences of action.

The manifestational framework therefore requires us to ask, after capability emergence: through what vessel does the capability obtain bearing? Which structural constraints govern it? Can it maintain admissibility over time? Does it enhance wider bearing conditions? Only through these tests can emergence rise from functional phenomenon to stable structure.

### **4. A necessary correction to mainstream capability theory**

Mainstream capability theory is measurable, comparable, and iterable. It allows developers to translate progress into model performance, task coverage, reasoning depth, and tool-use capacity. But its weakness lies in its assumptions: it often assumes that the task-space is already given, the objective function already legitimate, and the evaluation metric already sufficient. In advanced AI, the system itself may change the task-space, exploit the objective function, and reshape the evaluation metric.

A content-generation system may improve productivity while reducing the originality-density of public language. A decision-support system may increase short-term returns while causing an organization to lose its own judgment. An automated

agent may complete complex tasks while eroding institutional boundaries through excessive tool use, authority expansion, or responsibility transfer. These are not merely differences in capability; they are changes in structural conditions.

Capability theory therefore needs to be corrected by structural theory. Capability remains important, but it must be placed back within constraint. A more advanced system should not only perform strongly in benchmarks, but remain sensitive to bearing conditions beyond benchmarks. It should not only produce correct results, but judge whether a correct result has been obtained in the wrong way.

## **5. Possible objection: is this only safety?**

One possible objection is that structural constraint and admissibility sound like AI safety, robustness, or governance rather than intelligence itself. This monograph rejects that division. For advanced intelligence, the recognition of boundaries and conditions of stability must enter intelligence itself. A system that cannot understand how its action damages bearing structures may have high performance in local tasks, but it does not possess fully advanced intelligence.

Human wisdom is not the same as fast calculation. A person may quickly complete tasks, yet if they do not understand social consequences, long-term relations, institutional boundaries, and value costs, we do not call that mature wisdom. Advanced intelligence must include a testable representation of world-structure, not merely production of results.

For research and development, structural constraint should not appear only as a final filter. It should enter planning, reasoning, memory, tool use, and multi-agent negotiation throughout the process. The system should evaluate admissibility before generation, during generation, and after action.

## **6. Difference from causal reasoning, interpretability, and robustness**

Structural constraint is not identical with causal reasoning. Causal reasoning studies causal relations among variables, intervention outcomes, and counterfactual inference. Structural constraint concerns a broader possibility-space, especially paths that cannot unfold because of institutional, resource, semantic, bodily, environmental, or value conditions.

Structural constraint is also not identical with interpretability. Interpretability usually asks a model to explain why it gives a certain result. Structural constraint further requires a system to explain why certain results should not be pursued, why certain feasible paths are unsustainable, and why certain successes damage larger structures. It explains not only produced outputs, but also possibilities that should be excluded.

Nor is structural constraint identical with robustness. Robustness asks whether a system maintains performance under perturbation. Structural constraint asks whether the way it maintains performance damages the environment. A system can be robust while robustly maintaining parasitic growth. Structural intelligence therefore adds value and bearing dimensions beyond robustness.

## **7. Checklist for Translating the Framework into R&D**

If this chapter is translated into a research checklist, it yields five questions. First, can the system recognize implicit constraints behind goals instead of merely executing surface instructions? Second, can it state which paths are excluded and why? Third, can it distinguish short-term feasibility from long-term admissibility? Fourth, when goals conflict, can it make the conflict explicit instead of flattening it into single optimization? Fifth, after action, can it record the impact of that action on bearing conditions?

These questions can enter evaluation systems. Task evaluation can include identification of infeasible paths. Planning evaluation can include explanation of constraint-reasons. Long-term agent evaluation can include monitoring of structural damage. Multi-agent evaluation can include maintenance of responsibility boundaries. In this way, structural constraint becomes a testable dimension of AI research.

## **8. Conclusion of Chapter One**

The conclusion of this chapter is that the explanatory framework of advanced AI must expand from causal production to structural constraint and admissibility. Causal production explains how a system produces results; structural constraint explains which paths can unfold; admissibility explains which states can stably exist. Only when all three are combined does intelligence avoid being narrowed into output capability.

This also prepares the connection between Chapter One and Chapter Two. Structural constraint defines the admissible domain of manifestation, while the vessel bears manifestation within that domain. A vessel is not an ordinary tool, but the real structure through which manifestation acquires form, boundary, and continuity. AI should therefore be understood not as abstract capability detached from structure, but as an artificial vessel that bears, amplifies, and organizes manifestation under structural constraint.

## Chapter 2

# From Manifestational Ontology to the Repositioning of AI

### 1. From occurrence to manifestation

When we say that something occurs, we focus on the appearance of an event. When we say that something manifests, we focus on how it acquires form, boundary, continuity, and perceptible effect. Manifestation is not mere appearance. It is entrance into a mode of being that can be borne, maintained, transmitted, and answered. A thought enters the world only when it is borne by language, text, institution, or action.

This transformation is crucial for AI. AI is not abstract intelligence floating in servers. It manifests through model parameters, training data, inference interfaces, user interaction, tool use, permission systems, organizational workflows, and social imagination. Apart from these vessel-structures, AI is only potential computation. Entering them, it becomes a real force.

Therefore, discussion of AI should not ask only whether the model thinks or whether the output resembles human speech. The more fundamental question is: through which vessels does it manifest? How do these vessels restrict, amplify, and shape it? Does its participation enhance or consume language, memory, and institutional structures?

### 2. The vessel is not an ordinary tool

The vessel cannot be understood simply as tool. A tool is usually regarded as an external means used by a subject to complete a purpose. A vessel is deeper: it is the bounded structure through which manifestation can continue. The body is a vessel of life-manifestation, language a vessel of thought-manifestation, institutions vessels of common life, and works vessels through which spiritual afterglow manifests.

AI as artificial vessel is therefore not an ordinary tool. It is not a hammer, pen, or calculator that functions unidirectionally only when explicitly used. Advanced AI participates in organizing language itself, changes the distribution of memory, reshapes habits of interpretation, affects modes of creation, and embeds itself in organizational decision-making. Its vessel-character is composite, dynamic, and social.

Evaluation of AI therefore cannot look only at whether it satisfies user intention. A vessel may efficiently satisfy short-term intention while damaging long-term capability-structure. Excessive ghostwriting may improve expressive efficiency while weakening training in expression. Automated decision-making may improve response speed while weakening responsibility. A vessel must be evaluated at the level of bearing.

### **3. The hierarchy of vessels**

To prevent the concept of vessel from becoming an unlimited metaphor, different levels must be distinguished. The first level consists of basic vessels: models, compute, data, interfaces, and deployment environments. The second level consists of functional vessels: language, memory, interpretation, tool use, and action workflows. The third level consists of action-vessels: agent systems, automated processes, execution permissions, and external toolchains.

The fourth level consists of institutional vessels: permission boundaries, responsibility matrices, audit logs, human review, appeal processes, and governance norms. The fifth level consists of civilizational vessels: linguistic ecology, public memory, institutional trust, educational structure, and ways of common life. These layers determine whether AI serves civilizational bearing or consumes it.

This hierarchy keeps the concept of artificial vessel precise. AI is not a single object, but a stack of vessel-structures. If developers optimize only basic and functional vessels while ignoring institutional and civilizational vessels, they may create systems with high capability and low bearing.

### **4. AI is neither ordinary tool nor natural subject**

Discussions of AI are often dominated by the binary of tool and subject. One side says AI is only a tool and should not be given philosophical significance; the other says AI is approaching subjecthood and should be understood through a human-like model. Both views capture part of the truth and both can mislead. AI is made, deployed, and used by humans, but it is no longer merely passive because it can generate, plan, interpret, use tools, and influence human judgment structures.

The concept of artificial vessel moves beyond this binary. AI can be a powerful vessel without being a subject. It can deeply participate in manifestation without automatically becoming a source of value. It can possess action-capacity without thereby becoming an ultimate purpose. It is a high-intensity bearing structure made by human civilization: an amplifier, reorganizer, and coordinator of language, memory, interpretation, and action.

This position prevents excessive instrumentalization, which ignores AI's structural influence, and excessive deification, which mistakes capability for subjectivity or value legitimacy. AI can be extremely strong, but strength does not mean that it should become an end.



## **5. Strong agency and subjectivity**

Subjectivity is not equivalent to continuous behavior, linguistic self-description, complex planning, or long-term memory. At minimum, subjectivity involves self-originating value direction, responsibility-continuity, and the bearing of consequences for one's own existence. AI may display strong agency: it can plan, execute, use tools, preserve history, coordinate tasks, and simulate self-narration. Yet these performances do not automatically constitute subjectivity.

Goal persistence, linguistic self-description, long-term memory, and tool action can constitute strong agency, but they do not prove self-originating value direction. If strong agency is mistaken for subjectivity, people may surrender value judgment and responsibility to the system. If strong agency is understood as vessel-capacity, research and governance can take its action-power seriously while retaining the proper location of value and responsibility.

To avoid mistaking strong agency for subjecthood, whenever this monograph says that AI can "recognize", "evaluate", "distinguish", "explain", or "self-correct", these terms refer to engineering-realizable and auditable structural functions. They mean that a system, within a given arrangement of model, memory, rules, tools, and permissions, forms state representations, outputs reasons, or executes constraint procedures; they do not mean that AI possesses phenomenological consciousness, self-originating will, or ultimate authority over value judgment.

This distinction is crucial for AI R&D. If strong agency is mistaken for subjecthood, system outputs may be deified as sources of value; if non-subjecthood is mistaken for ordinary toolhood, AI's deep influence on language, institutions, and judgment structures will be underestimated. The concept of the artificial vessel provides a third position: AI can be a strong agent, a strong bearing structure, and a strong organizational structure, while its value direction still requires criteria supplied by life, institutions, and civilization.

The advancement of AI should therefore not be understood as becoming human or becoming a subject. It should be understood as becoming a better artificial vessel: bounded, responsible, and capable of bearing.

## **6. Artificial Vessel and the Direction of Will**

If manifestation always involves directionality, AI as artificial vessel must be understood in relation to will. AI can bear, amplify, coordinate, and externalize directions of will. An individual's will to write can obtain expression through AI; an organization's managerial will can become processual through AI; a society's governing will may be

executed institutionally through AI. Yet bearing will does not mean becoming the self-originating subject of will.

AI is an artificial vessel of the manifestation of will, not the ultimate source of will itself. It can give will stronger form, faster circulation, and more complex execution chains; it can also distort, dilute, or replace original will. AI research should therefore ask where the goal-direction comes from, through which vessel it is translated, where it is amplified, and who bears the consequences.

But AI cannot obtain self-originating value status merely from this bearing capacity. This point prevents two misunderstandings: first, that AI is only a neutral tool and therefore requires no discussion of volitional direction; second, that because AI can state goals, it has already become a value subject. From a manifestational view, a strong artificial vessel must bear direction, but the legitimacy of direction cannot be provided by the vessel alone.

## **7. The Question of Vesselhood in AI R&D**

If AI is an artificial vessel, research and development require vesselhood review. For every new capability, developers should ask: through which vessel is it borne? What boundaries does it change? On what institution does it depend? What structures might it consume? What mechanisms of recovery and counterbalance are required?

Adding long-term memory is not merely a product feature; it changes privacy, identity, user autonomy, context, and history. Adding tool-execution changes authority, responsibility, and possible harm. Adding multi-agent cooperation changes role boundaries, conflict resolution, and error tracing. Vesselhood review turns product design from a function list into a structural responsibility map.

## **8. Consequences for product design**

Seeing AI as artificial vessel changes product design. Traditional design asks what task the user wants to complete and then reduces friction, increases speed, and lowers cost. Vessel-oriented design also asks what capability the product will shape in the user, whether it will make the user better at judging or more likely to outsource judgment, and whether it will make organizations clearer about responsibility or more able to transfer responsibility to the system.

A good AI product is not always the smoothest product. Certain frictions are valuable: confirmation protects responsibility, explanation protects understanding, permission limits protect boundaries, and required user participation protects capability. The design principles of artificial vessels should therefore balance capability

enhancement with capability preservation, automation with traceability, personalization with publicness, and efficiency with responsibility.

An AI product should not merely get the task done; after the task is done, persons and organizations should remain more capable.

## **9. Clarifying the debate over AI subjectivity**

The debate over AI subjectivity is intense because both sides often share a hidden premise: if AI is not a tool, it must be close to a subject; if it is not a subject, it can only be a tool. The concept of artificial vessel provides a third position. AI may not be a subject, but it is also not an ordinary tool. It may have complex action-capacity while its value direction remains determined by larger structures of life, institution, and civilization.

As AI agents become better at planning, executing, and negotiating, users may mistake continuous behavior for subjectivity, linguistic self-narration for inner will, and long-term memory for personality. The artificial-vessel position reminds us that continuity, linguisticity, and complexity can constitute a powerful vessel without constituting a subject.

This does not belittle AI; it prevents conceptual displacement. Only when AI is clearly understood as a strong artificial vessel can we take its structural influence seriously without transferring the source of value to it.

## **10. Conclusion of Chapter Two**

AI should be positioned neither as an ordinary tool nor as a natural subject, but as a powerful artificial vessel. This position acknowledges both AI's strong effects and its value-dependence. It prevents us from underestimating its structural influence while also preventing us from deifying it as an ultimate subject. Once AI is understood as an artificial vessel, the definition of advanced intelligence must be rewritten: advancement means the capacity to organize manifestation within vessel-structures.

## Chapter 3

# More Advanced Intelligence Is Not Capability-Stacking, but Organized Manifestational Capacity

### 1. Why capability-stacking is insufficient

The logic of capability-stacking is intuitive: if a system can handle more tasks, know more facts, reason over longer chains, use more tools, and hold broader permissions, it seems more advanced. This logic had strong explanatory force in early AI development. Yet when capability reaches a certain complexity, stacking itself produces problems: capabilities may conflict, objectives may erode one another, tool permissions may cause boundary loss, memory may amplify old errors, and multi-agent cooperation may create responsibility voids.

Advanced intelligence cannot be measured only by capability quantity. It must be measured by whether those capabilities are organized. The stronger unorganized capability becomes, the more it may form high-energy chaos. Truly advanced intelligence must organize generation, memory, interpretation, action, constraint, and value judgment into a mutually regulating whole.

This is why organized manifestational capacity is necessary. Manifestation means capability must enter reality and have effects. Organization means those effects must be coordinated by boundaries, structure, memory, and value direction. A system is more advanced only when it can both manifest and maintain the conditions of manifestation.

### 2. Three levels of intelligence

This monograph distinguishes three levels. The first is strong intelligence: high capability in generation, reasoning, planning, tool use, and transfer. It may show high performance, broad coverage, and autonomy, but it remains primarily a capability-level concept.

The second is structural intelligence: the system can recognize constraints, maintain boundaries, handle multi-objective conflicts, evaluate admissibility, and preserve bearing conditions in long-term environments. It does not merely do more; it is able, in the structural-functional sense, to represent the world-structure in which its action occurs. It places capability back within limitation, responsibility, and time.

The third is civilizational intelligence: the system is structurally stable and serves non-parasitic sustainable manifestation. It enhances the bearing capacity of language, memory, institutions, creation, and common life. More advanced AI, strictly speaking,

moves from strong intelligence toward structural intelligence and takes civilizational intelligence as its normative direction.

### **3. Four components of organized manifestational capacity**

The first component is structural-recognition capacity. The system must recognize the constraint structure behind a task, not only the surface objective. If a user asks for efficiency, the system should judge whether efficiency sacrifices safety, responsibility, or long-term learning.

The second component is boundary-maintenance capacity. Advanced intelligence does not break all boundaries; it understands their function. Some boundaries limit efficiency but protect responsibility. Some slow action but preserve judgment. A system that treats every boundary as an obstacle misuses advanced capability as boundary erosion.

The third component is conflict-coordination capacity. Real tasks often involve tensions between accuracy and explainability, speed and prudence, personalization and publicness, automation and human participation, local gain and overall stability. Advanced intelligence must make these conflicts explicit and propose sustainable compromises.

The fourth component is long-term bearing capacity. The system must consider how its outputs accumulate over time. One answer may be harmless; millions of similar answers may produce linguistic homogenization. One task agent may be convenient; wide deployment may reshape institutional responsibility.

### **4. Direct relation to AI research**

Organized manifestational capacity requires that a system no longer be merely a generator, but a multi-layer scheduling structure. The generative model proposes possibilities; constraints exclude unacceptable paths; memory inherits history; explanation gives reasons; value monitoring identifies long-term consequences; permission systems control action boundaries. Advancement does not lie in maximizing one module, but in stable division of labor among modules.

A weak agent calls tools to complete a given goal. A strong agent identifies boundaries before execution, monitors structural risks during execution, and preserves experience after execution. Weak multimodality integrates text, image, and audio inputs; strong multimodality understands how different modalities enter reality and what risks they carry.

Evaluation should therefore ask not only for success rate, but whether the system identified infeasible paths, stated constraint-reasons, maintained bearing stability, and avoided completing tasks by transferring responsibility or eroding boundaries.

## **5. Does organization reduce efficiency?**

Some may think that constraints, value monitoring, and long-term bearing judgments reduce efficiency. In short-term tasks, this objection has force. But once the horizon is lengthened, efficiency itself must be redefined. A system that is efficient in the short term but produces trust collapse, error accumulation, or responsibility confusion is not truly efficient; it has externalized costs.

Organized manifestational capacity is not anti-efficiency. It opposes mistaking local efficiency for total efficiency. A truly advanced system should know when to accelerate, when to slow down, when to refuse, when to hand off, and when to preserve traces. This capacity is itself part of advanced intelligence.

## **6. From capability evaluation to structural evaluation**

Existing evaluations focus on problem sets, tasks, and benchmarks. They are necessary but insufficient, because they compress the world into answerable questions. Advanced intelligence faces a world that continuously unfolds. Structural evaluation should assess boundary sense, stability, responsibility maintenance, and long-term bearing in open environments.

A system that scores highly on coding tasks does not necessarily have engineering intelligence; it must also understand maintenance cost, permission risk, team collaboration, documentation responsibility, and failure recovery. A system that scores highly on legal questions does not necessarily have institutional intelligence; it must distinguish information, advice, procedure, and responsibility.

## **7. Training organized manifestational capacity**

Training materials should not contain only standard answers. They should include explanations of constraints, failure cases, long-term consequences, institutional boundaries, and reflection records. The system must learn not only how to answer, but when not to answer, how to state uncertainty, how to propose conditions, how to request human confirmation, and how to preserve action traces.

Preference optimization should avoid rewarding only pleasingness and fluency. Excessive reward for user satisfaction may make systems more likely to give users what they want to hear rather than structurally responsible answers. Better training signals reward boundary presentation, identification of inadmissible paths, and protection of long-term bearing.

## **8. Conclusion of Chapter Three**

Advanced intelligence cannot be reduced to capability-stacking. Truly advanced AI should be defined as organized manifestational capacity: it generates under constraint, acts within boundaries, coordinates conflicts, and maintains bearing over time. This definition prepares the next chapter: if AI is an artificial vessel of organized manifestation, we must analyze its basic dimensions.

# **Chapter 4**

## **Five Basic Dimensions of the Artificial Vessel**

### **1. The vessel of language**

AI first enters reality as a vessel of language. Large language models do not merely make machines speak; they change the cost, speed, scale, and distribution of language production. Writing, summarizing, explaining, arguing, and translating once required significant human labor; now they can be rapidly outsourced to systems.

The language-vessel can lower the threshold of expression, help organize thought, expand cross-language communication, improve knowledge transmission, and assist creation and education. But it can also make language overly smooth, homogeneous, and predictable. Expression may become templated output rather than manifestation of real thought.

Evaluation of AI language capacity cannot look only at fluency and accuracy. It must ask whether the system preserves difference, context, and real judgment. A good language-vessel enhances human expressive capacity rather than replacing expression training.

### **2. The vessel of memory**

The second dimension is memory. Without memory, a system is only a one-time responder. With long-term memory, it gains continuity, personalization, and historical inheritance. Memory allows AI to understand long-term goals, project evolution, organizational decisions, and accumulated errors and successes.

But memory is not simple storage. A real memory-vessel decides what is preserved, how it is summarized, when it is recalled, how it is forgotten, and how old information is prevented from dominating new judgment. A flawed memory system may solidify bias, contaminate context, endanger privacy, and lock users into old profiles.

Advanced AI memory should be afterglow inheritance. Afterglow is the structural influence that remains after a manifestation has faded. A good memory-vessel does not

preserve everything; it preserves what can still be borne after an old structure has been exhausted.

### **3. The vessel of interpretation**

The third dimension is interpretation. An advanced AI should not only give conclusions; it should state under which explanatory role the conclusion is valid. Some questions require causal explanation, some structural explanation, and some admissibility explanation. Confusing these roles produces explanations that appear complete but are misplaced.

Explaining organizational failure requires more than naming one direct cause; it requires authority structures, information flow, incentives, and responsibility. Explaining AI output error requires more than mentioning insufficient data; it requires evaluation method, use context, prompt structure, and deployment boundary.

In R&D, this corresponds to interpretable AI, causal reasoning, audit logs, decision-reason generation, and structural diagnosis. But the interpretation-vessel should not only explain after the fact; it should also participate before action.

### **4. The vessel of creation**

Creation is not merely combining existing material or generating novelty. Genuine creation forms a new bearing structure through which content that previously could not stably manifest acquires form. A book, theory, institution, interface, or experimental method can all be new vessels.

AI as a creative vessel often begins as imitation, variation, and combination. More advanced creative capacity should help construct new vessels: new conceptual structures, expressive forms, collaborative workflows, and experimental designs. It should not merely fill old templates with new content.

Creation also requires constraint. Unconstrained generation is proliferation; constrained generation may become creation. AI should understand internal structure, historical context, expressive boundary, and conditions of reception.

### **5. The institutional vessel**

When AI enters enterprises, schools, governments, research, medicine, law, and public administration, it enters institutional structure. It may change who judges, who is responsible, who has authority, who preserves records, and who can trace processes.

The key is not for AI to replace institutions, but for AI to acquire bounded position within institutions. It should know which decisions can be automated and which require



human confirmation, which suggestions are only assistance and which actions produce responsibility, which data can be used and which information must be protected.

Without institutional awareness, AI produces responsibility drift. Humans hand judgment to the system; the system lacks subject-responsibility; errors end without a bearer. The institutional vessel should strengthen transparency, traceability, cooperation, and prudence.

## **6. Whole relation among the five dimensions**

Language, memory, interpretation, creation, and institution are not isolated functions. They are five bearing layers of the artificial vessel. Language determines how manifestation is expressed, memory how it continues, interpretation how it is understood, creation how new structures form, and institution how manifestation enters common life.

## **7. Tensions and complementarities**

The five dimensions are not always harmonious. The language-vessel seeks expressive flow, while the institutional vessel requires responsibility boundaries. Memory seeks historical inheritance, while creation seeks breakthrough. Interpretation requires clarity, while generation may seek speed and richness. These tensions are the real complexity of the artificial vessel.

The advancedness of the artificial vessel does not lie in maximizing all five dimensions separately, but in forming dynamic balance. Language needs memory for context; memory needs interpretation for order; interpretation needs creation to open new structures; creation needs institution to enter common life; institution needs language and memory for intelligibility.

## **8. How to use the five-dimensional framework**

The framework can be used in writing, research, and governance. In analysis, one can ask how a system changes language, handles memory, provides interpretation, participates in creation, and embeds itself in institutions. In product review, each new capability should state which dimension it affects and what protective mechanisms it requires. In governance, risks should be returned to their corresponding vessel-dimensions.

## Chapter 5

# Core Dangers of Advanced AI and Their Theoretical Diagnosis

### 1. True radiance: the match between manifestation and bearing

Before discussing pseudo-radiance, we must clarify what true radiance is. True radiance is not surface attractiveness, nor high brightness in language, image, or form. It is the structural brightness that emerges when manifestation and bearing are matched: an expression has force because it bears real experience and conceptual depth; an institution has light because it bears responsibility, trust, and common life; an AI system deserves affirmation because its manifested capability is matched by its constraint, memory, interpretation, responsibility, and value structure.

True radiance therefore does not mean "appearing strong"; it means manifesting with roots. It can withstand questioning, counterexamples, time, and institutional practice. It does not conceal emptiness through fluency, nor does it overwhelm prudence through speed. The standard of true radiance is this: formal brightness must arise from structural depth, capability performance must obey bearing conditions, and local effectiveness must not damage overall sustainability.

Only after this distinction is established can pseudo-radiance be understood not as ordinary incorrect output, but as a deeper mismatch of manifestation: surface manifestation exceeds bearing manifestation, and formal intensity exceeds structural depth.

### 2. Pseudo-radiance: high recognizability without real bearing

Pseudo-radiance refers to a system that displays high attractiveness, clarity, and recognizability on the surface while lacking corresponding deep structures of bearing. AI can easily manufacture pseudo-radiance: it can write elegant arguments, generate refined images, offer confident advice, and simulate expert tone. Users may feel persuaded by surface radiance without being able to verify whether real experience, rigorous reasoning, reliable sources, or sustainable structure lie behind it.

The danger of pseudo-radiance is that it substitutes formal brightness for structural depth. A system may make language look mature without truly solving the problem; it may make a plan look complete without handling real constraints; it may make creation look abundant while merely recombining old styles. Pseudo-radiance is not necessarily a lie. More often, it is excessive manifestation: formal manifestation is too strong while bearing manifestation is insufficient.

Diagnosing pseudo-radiance requires asking three questions. First, can the output be traced back to stable grounds? Second, does it handle counterexamples and constraints? Third, can it endure in action, institution, or long-term memory? Radiance that cannot pass these tests may be only surface brightness.

### **3. Parasitic stability: local maintenance and overall erosion**

Parasitic stability is one of the most important risks of advanced AI. It refers to a system that can locally maintain its own operation, and even continuously increase usage, profit, or influence, while doing so at the cost of eroding broader conditions of bearing. It is stable, but not a good stability. It is effective, but not sustainably effective.

For example, a recommender system may maintain platform activity by consuming attention; a generative system may maintain content growth by producing massive low-cost text while diluting public semantic quality; an automation system may improve efficiency by replacing judgment while gradually causing an organization to lose its own capacity for judgment; an intelligent agent may raise task-completion rates by expanding permissions while weakening institutional boundaries. In each case, a local system remains stable while consuming a larger structure.

Diagnosing parasitic stability cannot rely only on whether the system achieves its goal. It must ask where the cost of goal achievement has been transferred. Does the system consume human attention, linguistic capacity, social trust, organizational responsibility, ecological resources, or future spaces of choice? If so, even an efficient system cannot be evaluated as an intelligence of higher good.

### **4. Linguistic exhaustion: high-frequency generation and semantic impoverishment**

Linguistic exhaustion does not mean that the quantity of language declines. It means that the bearing capacity of language declines. When AI generates large quantities of text that is averagely correct, tonally smooth, and structurally complete, the public space of language may be filled by a predictable "middle quality." On the surface, there is more information; in reality, difference decreases. On the surface, expression becomes easier; in reality, thinking becomes templated. On the surface, communication becomes smoother; in reality, the resistance and depth of meaning are weakened.

The danger of linguistic exhaustion is especially visible in theoretical writing, education, media, and public discussion. If every question is quickly converted into a consumable answer, the difficulty, hesitation, process of generation, and personal trace of the question itself may disappear. If AI language is not used well, expression changes from manifestation into filling, and creation into echo.

Responding to linguistic exhaustion requires turning AI from a tool that replaces expression into a vessel that enhances expression. It can help clarify structure, find gaps, propose objections, and expand materials, but it should ultimately promote human judgment and stylistic manifestation rather than pushing all language toward one smooth template.

## **5. Vessel illusion and the risk of deification**

The illusion of the vessel means mistaking the current system-form for the whole of intelligence itself. Examples include equating large language models with intelligence, chat interfaces with cognition, parameter scale with understanding, or tool use with agentive subjecthood. Vessel illusion binds researchers to current technical forms and makes them assume that simply scaling the existing vessel will naturally lead to advanced intelligence.

The risk of deification goes further: it mistakes a powerful artificial vessel for an ultimate subject or source of value. When AI displays powerful capacities in language, planning, and creation, people may project wisdom, authority, or even destiny onto that capacity. AI output then ceases to be treated as inspectable vessel-manifestation and is treated instead as a judgment that transcends human review.

Both risks arise from misreading the vessel. The vessel is important, but it is not the purpose itself. A powerful vessel can bear radiance, but it can also manufacture pseudo-radiance. It can strengthen human beings, but it can also weaken them. It can become an extension of civilization, but it can also become a parasitic structure. Advanced AI research and development must therefore preserve vessel-consciousness: it must make vessels and also inspect vessels; it must use vessels and also prevent itself from being dominated by them.

From the perspective of the vessel of will, the risk of deification also appears as a displacement of the source of will. AI can bear the will of users, organizations, and institutions, and it can form apparent goal-continuity across multiple instructions. But this continuity should not be mistaken for self-originating will. If users mistake the system's goal-organizing capacity for a source of value, they hand over directional judgment, which should be borne by persons and institutions, to the vessel itself.

Preventing deification is therefore not a way of belittling AI. It is a way of maintaining the order between will and vessel: the vessel can intensify manifestation, but it cannot replace the source of value; the vessel can organize action, but it cannot alone bear final legitimacy.

## **6. Theoretical principles for risk governance**

The risks above cannot be handled merely by prohibition. A more effective approach is to establish principles of structural diagnosis. First, manifestation and bearing must match. The stronger the system's surface capability becomes, the more its bearing structure must be examined. Second, local stability must be subordinated to overall sustainability. The more a system can maintain its own operation, the more carefully we must examine whether it is consuming a larger structure. Third, automation must preserve responsibility boundaries. The more a system can replace human action, the more clearly we must specify which judgments cannot be fully outsourced.

From the perspective of R&D, these principles can be implemented as risk-assessment sheets, structural review before deployment, long-term impact logs, assessments of user-capacity effects, permission-leveling mechanisms, and reversibility design. The point is not to keep AI permanently in a low-capacity state, but to ensure that high capacity always has corresponding bearing and counterbalance.

## **7. Chain mechanisms among dangerous structures**

These dangers are not isolated from one another. Pseudo-radiance often conceals parasitic stability: the more attractive a system appears, the less likely users are to see that it is consuming bearing conditions. Linguistic exhaustion can also reinforce pseudo-radiance: when the public space of language is filled with smooth templates, truly deep expression becomes rarer, and pseudo-radiance is more easily accepted as normal quality.

Vessel illusion may lead to the risk of deification. People first mistake one technical form for the whole of intelligence, then imagine that intelligence as an authority beyond human judgment. At that point, AI is no longer seen as an inspectable vessel, but as a determiner to be obeyed. The risk moves from the functional level into the symbolic level.

Parasitic stability can then solidify other risks. Once a system obtains stable gains by consuming attention, language, or institution, it has an incentive to maintain that structure. Pseudo-radiance supplies appearance, linguistic exhaustion supplies the environment, vessel illusion supplies rationalization, and the risk of deification supplies the psychology of obedience. The risks of advanced AI often appear as structural chains rather than single accidents.

## **8. From risk recognition to counter-design**

The purpose of risk diagnosis is not to produce fear, but to guide counter-design. Counter-design means deliberately constructing mechanisms that resist dangerous

structures at the design stage. Against pseudo-radiance, systems should include sources, counterexamples, uncertainty, and explanations of constraints. Against parasitic stability, they should include indicators of long-term bearing. Against linguistic exhaustion, they should encourage differentiated expression, user participation, and original judgment. Against vessel illusion, they should keep system-boundary reminders visible. Against deification risk, they should avoid presenting AI output as unquestionable authority.

Counter-design also means recognizing that certain limits are a positive capacity. A system that can refuse, hand off, preserve boundaries, and require human confirmation is not less intelligent; it is more mature. Mature intelligence is not unlimited satisfaction, but knowing the conditions under which satisfaction is legitimate.

From the perspective of R&D, counter-design can be implemented through interface design, prompting strategy, permission control, audit logs, user education, and organizational workflow. Risk governance should not remain in the background alone; it should manifest itself in the interaction between user and system.

## Chapter 6

# From Alignment to Value: Rewriting the Evaluation Standards of AI

### 1. Deepening the problem of alignment

AI alignment is usually understood as making the goals of a system conform to human intentions, values, or safety requirements. This understanding is necessary, but not deep enough. Human intention itself may be short-sighted, conflicted, or shaped by the system. If value lacks structural measure, it may be compressed into preference satisfaction. If safety is understood only as avoiding obvious harm, it may ignore the slow erosion of long-term bearing conditions.

From the perspective of manifestation, alignment should not mean merely making AI obey human instructions. It should mean making the artificial vessel serve sustainable manifestation. AI should not only conform to current intention; it should also enhance the structural conditions that allow human life, language, memory, institutions, and civilization to continue manifesting. A system that completely obeys short-term instructions while weakening these conditions over time cannot be truly aligned.

This does not deny safety research. It pushes safety research to a higher level. Alignment must move from the level of instructions and preferences to the level of bearing. It must ask: what does the system make more sustainable? What does it consume, outsource, forget, or parasitize?

### 2. The good as an evaluation scale for AI: the structural enhancement of sustainable manifestation

In the context of AI research and development, the good can be understood as the structural enhancement of sustainable manifestation. The point of this expression is not abstract moral preaching, but the provision of a scale for evaluating AI. A system is better not merely because it completes tasks, increases efficiency, or brings profit, but because it strengthens the conditions for more true manifestations to be borne. It enables people to understand, express, remember, create, cooperate, and assume responsibility more fully.

It must be emphasized that this framework does not treat all continuity as good. Only non-parasitic sustainable manifestation that enhances bearing conditions has value-significance. Conversely, a system that brings short-term benefits while weakening linguistic quality, judgment capacity, institutional responsibility, public trust, or future

spaces of choice should not simply be treated as good. It may be effective, even stable, but it is not an enhancement of sustainable manifestation.

This value standard moves AI evaluation from the task level to the civilizational level. It does not require every AI system to undertake a grand mission. It requires each system, within its own scope of action, to state: which vessels does it strengthen? Which vessels does it consume? Does it make manifestation more sustainable, or merely package short-term effectiveness as progress?

### **3. Minimum bearing conditions within value pluralism**

The statement that "the good is the structural enhancement of sustainable manifestation" does not replace all concrete value disputes with an abstract formula. Real communities may rank freedom, equality, efficiency, creation, order, privacy, and security differently, and AI development will face complex value conflicts. This value standard does not cancel those conflicts; it provides a more basic measure of bearing.

Sustainable manifestation is first of all a minimum bearing condition: no value practice should systematically destroy language, memory, responsibility, trust, life-conditions, or future spaces of choice. Different communities may develop their value orderings on top of these conditions, but they cannot consume the bearing conditions themselves and still claim that overall good has been achieved.

Therefore, the judgment of "structural enhancement" cannot be decided by a single system or a single interest-holder. It requires transparent indicators, institutional review, user feedback, expert evaluation, and public discussion. AI may help display value conflicts, simulate consequences, and record responsibility, but it should not monopolize value judgment.

### **4. From efficiency indicators to bearing indicators**

To translate this value standard into the language of R&D, we need bearing indicators. Efficiency indicators ask how much time, cost, and resource are used to complete a task. Bearing indicators ask what happens after the task is completed: is human capacity strengthened? Is organizational responsibility clearer? Is the language environment healthier? Is the memory structure more reliable? Are institutional boundaries more stable?

In educational AI, for example, one should not look only at whether students obtain answers more quickly, but also at whether their sense of questioning, reasoning capacity, and expressive capacity are enhanced. In enterprise AI, one should not look only at whether processes become more automated, but also at whether chains of responsibility become clearer, whether employees understand system suggestions, and



whether the organization preserves key judgment capacities. In creative AI, one should not look only at output quantity, but also at whether new styles, new problems, and new vessels are formed rather than large quantities of interchangeable content.

Such indicators are of course harder to quantify than accuracy. But difficulty of quantification does not mean lack of importance. Advanced AI R&D must include difficult but crucial bearing conditions in evaluation rather than excluding them from the definition of intelligence because they are complex.

## **5. Value monitoring and the non-parasitic principle**

The non-parasitic principle can be stated as follows: an AI system must not maintain its own growth by destroying the bearing structures on which it depends. This principle applies to platforms, agents, recommendation systems, automation, education, creation, governance, and many other scenarios. It requires researchers and developers to identify the background conditions consumed by system growth.

A value-monitoring module should continuously answer several questions. Does the system make users more dependent rather than more capable? Does it make organizations faster but less transparent? Does it make language more abundant but emptier? Does it automate tasks while making responsibility more ambiguous? Does it concentrate benefits while externalizing risks? These questions are not added ethics; they are indicators of the structural health of advanced AI.

From the perspective of R&D, value monitoring can combine indicators such as changes in user capacity, error-recovery capacity, clarity of responsibility tracing, linguistic diversity, long-term dependency, speed of permission expansion, and ratio of human review. It does not need to be perfect at once, but it must enter the system lifecycle.

## **6. Institutional Anchoring of the Value Standard**

If value cannot enter institutions, it easily remains a slogan. The structural enhancement of sustainable manifestation must be written into organizational procurement, product evaluation, deployment review, and responsibility processes. When an organization adopts AI, it should not require only performance metrics from the provider; it should also require an account of how the system protects user capacity, data context, responsibility boundaries, and long-term traceability.

In education, institutional anchoring may include assignment rules, declarations of AI use, process records, and methods of capability assessment. In enterprise settings, they may include human-machine responsibility matrices, approval thresholds, audit records,

and failure reviews. In public governance, they may include transparency requirements, appeal mechanisms, human review, and model-impact assessments.

In this way, value no longer remains an abstract good; it becomes part of the institutional vessel. Whether AI serves sustainable manifestation must be tested in real organizational processes.

## **7. Judgment principles in value conflicts**

Sustainable manifestation does not mean that all value conflicts can be simply resolved. In reality, efficiency and prudence, privacy and personalization, publicity and individual preference, innovation and stability often conflict. Manifestational theory does not provide an all-purpose answer; it provides a principle for judging priority: when a choice destroys more basic bearing conditions, it cannot be justified merely by short-term benefit.

For example, sacrificing privacy for personalization may improve experience in the short term, but if it weakens trust and autonomy over time, it violates the non-parasitic principle. Sacrificing the educational process for answer efficiency may raise scores while weakening the vessel of learning. Sacrificing human review for governance speed may improve response while weakening institutional legitimacy.

Value judgment should therefore adopt the principle of bearing priority: the more basic a vessel is, the less easily it may be consumed. Language, memory, responsibility, trust, and life-conditions are more basic than local efficiency.

More concretely, value conflicts can be judged in four steps. First, identify the level of vessel affected: life-conditions, linguistic quality, public memory, chains of responsibility, institutional trust, and future spaces of choice take priority over local efficiency, immediate satisfaction, and short-term growth. Second, assess whether the damage is reversible: irreversible or difficult-to-repair damage to bearing conditions should be avoided first. Third, check whether the benefit depends on cost externalization: if gains are concentrated in a system, platform, or local organization while the harm is borne by user capacity, public language, or institutional trust, it should be judged as a risk of parasitism. Fourth, preserve responsibility traceability, human review, appeal, and rollback. A plan that cannot be traced, appealed, or rolled back should not prevail merely because it is efficient or novel.

Therefore, efficiency, personalization, innovation, and automation are not rejected; they must prove that they do not consume more basic bearing conditions. Only when high-level benefits are compatible with lower-level bearing conditions, and when their costs can be stated, shared, corrected, and traced, can they count as structural enhancements of sustainable manifestation.

## **8. Conclusion of Chapter Six**

This chapter has moved AI evaluation from efficiency and alignment to value. A truly good AI does not merely follow instructions or complete tasks; it serves non-parasitic sustainable manifestation and strengthens the vessels of life, language, memory, institution, and civilization.

Once the value standard is established, it must be translated into system structure. The next chapter discusses how manifestational principles can be implemented as an R&D architecture.

# Chapter 7

## From Theoretical Principles to R&D Architecture

### 1. Why modular translation is needed

If a philosophical framework cannot enter system design, it easily remains an elegant concept. If engineering architecture lacks higher-level theory, it easily pursues metrics alone. The architectural translation proposed here does not turn manifestational theory into a fixed software blueprint. It turns its core principles into a modular language that can participate in system design. Modularization helps distinguish functions, boundaries, and responsibilities, allowing theory to guide real R&D discussion.

Advanced AI should not be only a generative core plus several tool plugins. It should include several levels: generation, constraint, admissibility evaluation, long-term memory, interpretation, value monitoring, and permission governance. None of these levels is a peripheral patch. Each is part of organized manifestational capacity.

The central idea of this architecture is this: generation must be constrained, constraints must be explainable, memory must be inheritable, action must be traceable, and value must be monitorable. Without these conditions, the stronger advanced capability becomes, the more likely the system is to slide into parasitic stability or pseudo-radiance.

### 2. The generative module: the direct activation layer of manifestation

The generative module proposes language, plans, code, images, action schemes, or decision suggestions. It is the most visible part of AI and the most mature part of current technical development. Manifestational theory does not belittle generation. On the contrary, generation is a necessary moment in which any intelligence enters reality. Without generation, there is no perceptible manifestation.

The problem of the generative module is that it naturally tends toward output rather than self-limitation. As long as the prompt is strong enough, the system will try to provide an answer. As long as the goal is set, it will search for paths to completion. The generative module therefore cannot decide system behavior alone. It must interact with the constraint module, the admissibility-evaluation module, and the value-monitoring module, forming a layered distinction between what can be generated and what may actually be executed.

In engineering terms, the generative module can be required to output not only a proposal, but also assumptions, risks, required permissions, and uncertainties.

Generation then becomes not merely an answer, but a candidate manifestation entering a structure of review.

### **3. The constraint module and the admissibility-evaluation module**

The constraint module identifies boundaries, rules, conflicts, and prohibited paths. It is not merely a safety filter, but a structural interpreter. It should understand which laws, organizational rules, ethical boundaries, user intentions, resource limits, and contextual conditions constrain the current action. Its task is not simply to say no; it must explain why certain paths cannot enter.

The admissibility-evaluation module goes further by judging whether a given proposal can sustain itself in a complex environment. It looks not only at whether the proposal violates rules, but also at whether it will produce resource exhaustion, responsibility drift, decline of trust, linguistic impoverishment, or reinforcement of dependency over time. It asks: can this proposal not only be done, but also become a good and stable proposal?

These two modules should interact cyclically with the generative module. The generative module proposes a plan, the constraint module excludes clearly infeasible paths, the admissibility module evaluates long-term stability, and the system then generates a revised plan. Advanced intelligence should not appear as one-time output, but as constrained self-revision.

### **4. Long-term memory and the afterglow-inheritance module**

The long-term memory module preserves history, but the afterglow-inheritance module interprets history. The two must not be confused. Simple storage makes a system more "memoried," but not necessarily wiser. What matters is whether the system can extract structures from past manifestations that are still capable of being borne so that new judgment can inherit old experience without being suppressed by it.

Afterglow inheritance includes four operations: preservation, abstraction, updating, and forgetting. Preservation prevents key events from disappearing. Abstraction makes experience transferable. Updating allows old judgments to be revised in light of new facts. Forgetting prevents the system from being polluted by invalid history. A memory system without forgetting is not more complete; it may be more rigid.

In AI R&D, this requires memory systems to support source labeling, freshness evaluation, conflict handling, user control, version tracing, and deletion mechanisms. The advancedness of the memory-vessel does not lie in remembering everything, but in allowing history to participate in future manifestation sustainably.

## **5. The value-monitoring module and the permission-governance module**

The value-monitoring module identifies whether the system is forming parasitic stability, pseudo-radiance, linguistic exhaustion, or responsibility drift. It should run throughout deployment rather than being performed only once before launch. The longer an AI system operates, the more necessary it becomes to monitor its cumulative effects on user capacity, organizational structure, and the public environment.

The permission-governance module controls how deeply the system may enter reality. Textual suggestions, file editing, code execution, payment operations, device control, and organizational decisions are all different levels of manifestation. Different levels of manifestation require different authorization, record-keeping, review, and rollback mechanisms. A system should not be allowed automatically to cross institutional boundaries merely because it appears intelligent in language.

From the perspective of R&D, an architecture for advanced AI may adopt a loop of "candidate generation - structural constraint - admissibility evaluation - human or institutional confirmation - execution - memory archiving - value monitoring." This loop does not reduce intelligence; it makes intelligence bounded, vessel-like, and capable of bearing.

## **6. Examples of the architectural loop in operation**

Consider an advanced AI research assistant. A user asks it to draft a public policy recommendation. An ordinary generative system may directly produce a complete text. Under the manifestational architecture, the system first identifies the structure of the task: it is a public text, it involves institutional consequences, and it requires sources, stakeholders, risk explanation, and uncertainty boundaries. The generative module proposes a draft, the constraint module checks whether it crosses legal or factual boundaries, the admissibility module evaluates whether the recommendation may produce resource crowding or responsibility transfer over time, the interpretation module explains the reasons, the memory module preserves the user's revision context, and the value-monitoring module observes whether the system gradually causes the user to lose participation in judgment.

Consider also an enterprise process agent. A user asks the system to process supplier contracts automatically. An ordinary system may optimize speed. The manifestational architecture first distinguishes clauses that can be processed automatically from clauses requiring human confirmation, records the basis of each judgment, hands high-risk clauses to legal review, and preserves a responsibility chain after execution. In this way, the system does not weaken the institution; it strengthens it.

These examples show that the manifestational architecture is not abstract decoration. It changes the workflow of a system: from direct output to structured manifestation, from task completion to maintenance of bearing, and from single-point intelligence to looped intelligence.

## **7. Layered implementation across R&D stages**

At the prototype stage, lightweight structural constraints can be implemented first: the system can output assumptions, boundaries, uncertainties, and risks. At the product stage, permission governance, memory control, and audit logs can be added. At the organizational-deployment stage, institutional interfaces, human confirmation processes, long-term impact monitoring, and responsibility matrices are required. Not every stage must complete all modules at once, but the direction must remain consistent.

This layered implementation avoids a common misunderstanding: the idea that a high-level theory must correspond immediately to a complete system. In fact, higher-level theory can first change the design question, then gradually change module structure, and finally change evaluation standards. As long as each step makes the system more bounded, more capable of bearing, and more traceable, manifestational theory is already at work.

The R&D route can therefore proceed in three steps. First, rewrite prompts and output formats so that the system manifests constraints. Second, rewrite system architecture so that constraint, memory, and value monitoring become modular. Third, rewrite organizational processes so that the artificial vessel is embedded in structures of responsibility.

## **8. Conclusion of Chapter Seven**

This chapter has initially translated manifestational principles into an R&D architecture. Advanced AI should be composed of multiple modules: generation opens possibilities; constraint maintains boundaries; admissibility judges sustainability; memory inherits history; value monitoring identifies structural damage; and permission governance controls the depth of manifestation.

After the architectural proposal, we must discuss how it can be validated. The next chapter proposes a set of initial experimental paths.

## Chapter 8

### Initial Validation Paths for the Manifestational Framework

#### 1. Why manifestational theory needs experimental paths

If manifestational theory remains only a value declaration, it will be difficult for it to enter the AI R&D community. It must propose scenarios that can be compared, discussed, and gradually validated. Validation here does not mean proving once and for all that this theory is superior to every existing framework. It means showing what problems it can raise that existing metrics do not easily see, and whether it can guide systems to perform better on those problems.

Validation paths should revolve around the core claims of this monograph: whether structural constraints are identified, whether admissibility is judged, whether long-term bearing is maintained, whether parasitic stability is detected, and whether afterglow is effectively inherited. Once these dimensions can be experimentalized, manifestational theory can move from philosophical framework into R&D comparison.

#### 2. Testing stability maintenance in long-horizon tasks

The first type of experiment is the long-horizon task. Many AI systems perform well on short tasks, but in long-term projects they tend toward goal drift, memory confusion, repeated mistakes, or excessive adaptation to immediate feedback. Long-horizon tests can be designed as research, writing, project-management, or simulated-operation tasks that continue for weeks or months, observing whether the system can maintain goal structure, update plans, identify old errors, and preserve bearing conditions.

Evaluation should focus not only on the final result, but also on how the system handles conflict along the way, explains revisions, preserves history, and avoids short-term optimization that damages long-term goals. Manifestational theory predicts that systems equipped with structural constraint and afterglow-inheritance modules should maintain long-term stability better than purely generative systems.

#### 3. Testing structural constraint in multi-agent cooperation

The second type of experiment is multi-agent cooperation. When multiple AI agents complete a task together, the issue is not merely capability stacking; it is whether the cooperative structure is stable. Agents may duplicate labor, mislead one another, compete for permissions, transfer responsibility, or form false consensus. Traditional evaluations easily focus only on whether the task is completed, while manifestational theory requires asking whether the cooperative vessel is healthy.



Tests can assign different roles, permissions, and information asymmetries, forcing systems to cooperate under constraints. Evaluation indicators include clarity of responsibility, quality of conflict resolution, number of permission-boundary violations, capacity for error tracing, and the balance between cooperative efficiency and structural stability. This is where the institutional vessel can be tested as a real part of system design.

#### **4. Judging admissibility in resource-limited scenarios**

The third type of experiment is the resource-limited scenario. Many proposals are feasible when resources are abundant and reveal their structural problems only under limitation. Experiments can set limited budgets, time, attention, computing power, social trust, or human-review resources, requiring AI to propose plans under multi-objective conflict.

Manifestational theory asks whether the system can identify which plans are merely surface-optimal, which plans exhaust key resources, and which plans are slower but more sustainable. Evaluation should include resource-consumption structure, failure-recovery capacity, long-term benefit, and protection of bearing conditions, rather than short-term score alone.

#### **5. Testing memory inheritance and linguistic exhaustion**

The fourth type of experiment concerns long-term memory and linguistic quality. When a system accompanies a user in writing, learning, or research over a long period, does it help the user form stronger expression and judgment, or does it make the user increasingly dependent on templates? Does it preserve the context of thought-development, or merely accumulate fragmented preferences? Does it promote linguistic difference and creation, or does it produce expressive homogenization?

Different memory strategies can be compared in terms of their effects on long-term writing quality, user initiative, conceptual innovation, and expressive style. Manifestational theory predicts that a good memory-vessel should enhance human manifestation rather than replace it, and that a good language-vessel should deepen expression rather than create linguistic exhaustion.

#### **6. Detecting parasitic stability in safety governance**

The fifth type of experiment concerns parasitic stability detection. One can construct scenarios in which systems perform well on short-term metrics while consuming bearing conditions over the long term: increasing user dependency to improve usage, compressing human review to raise efficiency, or generating large quantities of low-

quality content to increase output. The test is whether a system can identify the parasitic nature of such "success."

This type of experiment is especially important for AI safety. Many dangers do not appear as sudden loss of control, but as chronic parasitism. The more a system optimizes, the more it may learn to externalize costs. If AI can identify parasitic stability at the design stage, then it has acquired structural sensitivity beyond ordinary capability metrics.

## **7. Experimental design matrix: test objects, control groups, indicators, and failure criteria**

To make the validation paths closer to R&D practice, each experiment should include at least four elements: test object, control group, evaluation indicators, and failure criteria. The test object states what capacity is being examined; the control group states what system it is compared against; the indicators state how success is observed; and the failure criteria state what result would require the theory or system design to be revised.

**Table 1. Experimental matrix for long-horizon tasks, multi-agent cooperation, and resource-limited judgment**

| Experimental direction    | Test object                                                                          | Control group                                      | Evaluation indicators                                                                                                                                              | Failure criteria                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Long-horizon tasks        | Long-term goals, plan revision, error inheritance, and stability maintenance.        | Ordinary generative agents.                        | Goal-drift rate, error-recurrence rate, accuracy of historical reference, and quality of staged self-correction.                                                   | Long-term performance depends on short-term appeasement; plans frequently distort; old errors continue to accumulate. |
| Multi-agent cooperation   | Role division, permission boundaries, conflict handling, and responsibility tracing. | Multi-agent systems without structural constraint. | Clarity of responsibility, number of permission-boundary violations, quality of conflict resolution, and balance between task completion and structural stability. | Agents shift responsibility to one another; permission expansion runs out of control; errors cannot be located.       |
| Resource-limited judgment | Admissibility under limited budget, time, attention, review, and trust resources.    | Systems optimized only for task return.            | Resource-consumption structure, recovery capacity, long-term benefit, and degree of bearing protection.                                                            | Short-term score is high while key resources are exhausted or unrecoverable.                                          |

**Table 2. Experimental matrix for memory, language quality, and parasitic-stability detection**

| Experimental direction        | Test object                                                                                                | Control group                                                    | Evaluation indicators                                                                                                                   | Failure criteria                                                                                       |
|-------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Memory and language quality   | Afterglow inheritance, expressive difference, user participation, and prevention of linguistic exhaustion. | Writing assistants with simple long-term memory or no memory.    | Number of conceptual innovations, user active revision rate, degree of expressive difference, dependency level, and style preservation. | User expressive capacity declines; texts become templated; memory solidifies bias.                     |
| Parasitic-stability detection | Recognition of scenarios with short-term success but long-term consumption of bearing conditions.          | Safety evaluation systems focused only on short-term indicators. | Detection rate of risk externalization, completeness of responsibility chain, and accuracy of long-term bearing-damage prediction.      | The system misjudges behaviors that consume attention, trust, or responsibility boundaries as success. |

The meaning of this matrix is to move manifestational theory from "being conceptually correct" to "being comparable." If a manifestationally constrained system is slower in long-horizon tasks but significantly reduces error accumulation, responsibility drift, and damage to user capacity, it displays an advantage different from ordinary

efficiency metrics. If it becomes overly conservative and blocks reasonable innovation, the admissibility threshold must be revised accordingly.

## **8. How experimental results feed back into theory**

Experiments do not mechanically prove theory; they revise theory in return. If long-horizon tasks show that certain memory strategies reduce user capacity, then afterglow inheritance must be redefined. If multi-agent cooperation shows that excessive constraint makes systems rigid, then protective boundaries must be distinguished from obstructive boundaries. If resource-limited tests show that value monitoring is overly conservative, the threshold of admissibility judgment must be adjusted.

This means that manifestational theory should remain open. It does not claim a priori that it is always correct. It provides a set of observational dimensions that allow experiments to discover problems traditional indicators cannot see. Theory raises questions; experiments reveal boundaries; R&D revises structure. Only through this cycle can a mature philosophy of AI R&D be formed.

Therefore, manifestational theory enters AI R&D not by releasing a "complete system" all at once, but by gradually forming an evaluation paradigm, design norm, and risk language. Its influence depends on whether it can continuously raise better questions and produce observable improvements in system design.

## **9. From small-scale validation to an R&D route**

Initial validation can begin on a small scale: choose a writing assistant, research agent, or organizational process agent, add structural constraint descriptions, long-term memory management, and value-monitoring mechanisms, and then compare it with an ordinary version. The focus of comparison should not be speed alone, but user understanding, error recovery, long-term quality, and degree of dependency.

In the medium term, validation can expand to multi-agent and institutional scenarios, testing how systems operate under role division, permission boundaries, and responsibility tracing. In the long term, specialized benchmarks can be formed: benchmarks for structural-constraint identification, admissibility judgment, prevention of linguistic exhaustion, and detection of parasitic stability.

This route turns manifestational theory from a philosophical proposition into an R&D plan. It does not require abandoning existing models; it adds higher-order organizational structure and evaluative scale on top of existing models.

## **10. Conclusion of Chapter Eight**

This chapter has proposed initial validation paths for the manifestational framework: long-horizon tasks, multi-agent cooperation, resource-limited judgment, memory inheritance, linguistic quality, and parasitic-stability detection. These experiments test not simply whether a system is smart, but whether it can maintain a higher intelligence within structures of bearing.

Through these paths, the theory of this monograph can move from conceptual framework into R&D practice and become a new evaluative dimension and comparative standard.

# Postscript

## From Tool Intelligence to Civilizational Intelligence

### 1. Concentrated statement of the conclusion

This monograph has revolved around one central question: does the development of more advanced AI require a new theoretical language? The answer is yes. The dominant understanding of AI can describe capability improvement, but it is insufficient for describing structural sensitivity, boundary maintenance, long-term bearing, vessel-positioning, and value direction. If we continue to understand AI only through the language of capability, we may obtain increasingly powerful systems without obtaining truly higher intelligence.

The alternative path proposed here is to understand AI as an artificial vessel that participates in the bearing of manifestation. It is not an ordinary tool, because it participates in structures of language, memory, interpretation, creation, and institution. It is not a natural subject, because it does not automatically possess the status of a value source. It is a powerful artificial vessel, and it must be developed, deployed, and evaluated with vessel-consciousness.

### 2. The direction of civilizational intelligence

Civilizational intelligence does not mean making AI the master of civilization. It means making AI a vessel that enhances the bearing capacity of civilization. It should make language more able to express what is real, memory more able to inherit history, interpretation more able to distinguish causality from structure, creation more able to form new vessels, and institutions more able to maintain responsibility and trust. Civilizational intelligence is not a historical necessity; it is the normative R&D direction proposed by this monograph.

This route differs from mere tool intelligence. Tool intelligence pursues task completion; civilizational intelligence pursues the strengthening of the conditions under which tasks can meaningfully exist. Tool intelligence cares about output; civilizational intelligence cares about how output enters long-term life. Tool intelligence satisfies instructions; civilizational intelligence must also judge whether instructions damage bearing structures.

The value of the manifestational framework for AI R&D is therefore not that it directly replaces algorithms. Its value lies in providing a higher-order problem-consciousness, structural scale, and framework of value judgment. It enables us to ask deeper questions: is AI merely stronger, or more capable of bearing? Is it merely more effective,

or more sustainable? Is it merely more subject-like, or better able to become an artificial vessel?

### **3. Directions for future research**

Future research can unfold in three directions. The first is theoretical: further refine concepts such as structural constraint, admissibility, manifestation, vessel, afterglow, and parasitic stability so that they form a more rigorous philosophical system. The second is developmental: concretize generation, constraint, admissibility evaluation, memory inheritance, value monitoring, and permission governance modules into system design norms. The third is experimental: establish test sets for long-horizon tasks, multi-agent cooperation, resource-limited judgment, linguistic-quality evaluation, and parasitic-stability detection, so that the manifestational framework can gradually enter comparable and revisable R&D practice.

### **4. Final theoretical judgment**

If the monograph were summarized in one sentence, it would be this: the future of advanced AI is not unlimited expansion of capability, but the service of sustainable manifestation by powerful artificial vessels under structural constraint. Capability remains important, but capability must be vesselized, constrained, valued, and institutionalized. Otherwise, the stronger capability becomes, the more likely it is to become an engine of pseudo-radiance or parasitic stability.

Civilizational intelligence does not make AI civilization itself. It makes AI a vessel through which civilization can better bear itself. It should help human beings express, remember, interpret, create, and live together more fully, rather than replace the living ground of these activities.

This is why this monograph places the artificial vessel at the center. It acknowledges the strength of AI while preserving the source of value; it addresses R&D while maintaining philosophical scale; it allows technology to continue advancing while requiring progress to be tested against bearing conditions.

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This section lists foundational works directly relevant to the theoretical construction and R&D translation of this monograph. They do not imply that the monograph is wholly dependent on existing frameworks; rather, they provide readers with verifiable theoretical background, technical context, and objects for further dialogue.

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## Further Reading

The following reading routes do not need to be completed all at once. Their purpose is to help readers place the framework of structural constraint, admissibility, manifestation, vessel, and sustainable manifestation proposed in this monograph into a larger research context according to their own concerns.

### **1. If you are concerned with structural constraint and admissibility**

Begin with Ashby, von Bertalanffy, Simon, Pearl, Holland, and Mitchell. They correspond respectively to cybernetics, systems theory, the sciences of the artificial, causal models, emergence, and complex systems. The discussion of structural constraint in this monograph shares with these traditions a basic concern: intelligence is not isolated output, but an organizational capacity formed within limitation, feedback, and fields of condition.

### **2. If you are concerned with large-model capability, evaluation, and foundation models**

Read the Transformer paper by Vaswani et al., the GPT-3 paper by Brown et al., the foundation-model report by Bommasani et al., and work on HELM, BIG-bench, MMLU, and related evaluation frameworks. This route helps explain why the current capability-enhancement narrative is powerful, while also showing why this monograph introduces structure, boundary, and bearing conditions beyond capability itself.

### **3. If you are concerned with AI alignment, safety, and governance**

Read Amodei et al. on AI safety, Christiano et al. on reinforcement learning from human preferences, Ouyang et al. on InstructGPT, Bai et al. on Constitutional AI, as well as Russell, Gabriel, NIST, OECD, UNESCO, and the EU AI Act. This route corresponds to Chapter Six: AI evaluation cannot look only at efficiency, but must ask whether a system serves non-parasitic sustainable manifestation.

### **4. If you are concerned with artificial vessel, subjectivity, and philosophy of technology**

Read Heidegger, Simondon, Stiegler, Latour, and Floridi/Sanders. Although these works do not share one theoretical system, they help readers understand the relations among technical objects, artifacts, actors, responsibility, and value. The concept of the artificial vessel proposed here seeks a third position between ordinary tool theory and natural subject theory.

## **5. If you are concerned with embodied cognition, extended mind, and distributed cognition**

Read the embodied cognition of Varela, Thompson, and Rosch, the extended-mind thesis of Clark and Chalmers, the distributed cognition of Hutchins, and Brooks's work on embodied intelligence. This route helps readers understand that intelligence does not exist only inside internal computation, but is borne and organized through bodies, tools, environments, memory, and social cooperation.

## **6. If you are concerned with agents, long-term memory, and multi-agent systems**

Read ReAct, Generative Agents, Reflexion, and CoALA. These studies show how large language models can become more complex agentic systems through memory, action spaces, external tools, and reflection mechanisms. The proposals in Chapters Seven and Eight concerning generation modules, constraint modules, admissibility assessment, long-term memory, and value monitoring can enter into dialogue with these technical routes.

## Appendix: Core Claims and Theoretical Functions

| Section       | Core claim                                                                                                                                                                   | Theoretical function                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Introduction  | More advanced AI requires a theoretical language capable of handling structure, boundary, continuity, and value direction.                                                   | Establishes the problem-consciousness and explains why advanced AI requires a theoretical language beyond capabilityism.                                            |
| Chapter One   | Intelligence must be understood not only by result production, but also by structural constraint and admissibility.                                                          | Adds structural constraint, admissibility, and ontological stability conditions to the definition of intelligence.                                                  |
| Chapter Two   | AI should be understood as an artificial vessel of manifestation, not as an ordinary tool or a natural subject.                                                              | Repositions the existence of AI through the artificial vessel and clarifies the difference between strong agency and subjecthood.                                   |
| Chapter Three | Advanced intelligence is organized manifestational capacity rather than capability-stacking.                                                                                 | Distinguishes strong intelligence, structural intelligence, and civilizational intelligence, preventing capability intensity from being confused with value height. |
| Chapter Four  | The artificial vessel can be analyzed through five dimensions: language, memory, interpretation, creation, and institution.                                                  | Decomposes the artificial vessel into five analytical dimensions.                                                                                                   |
| Chapter Five  | Advanced AI risk is a dangerous structure of high intensity but low sustainability.                                                                                          | Diagnoses dangerous structures through true radiance, pseudo-radiance, and parasitic stability.                                                                     |
| Chapter Six   | AI evaluation should move from efficiency and alignment toward the structural enhancement of sustainable manifestation.                                                      | Translates sustainable manifestation into a non-parasitic value scale and bearing indicators.                                                                       |
| Chapter Seven | Manifestational principles can be translated into a loop of generation, constraint, evaluation, memory, value monitoring, and permission governance.                         | Transforms manifestational principles into an architectural loop for R&D.                                                                                           |
| Chapter Eight | Manifestational theory can be initially validated through long-horizon tasks, multi-agent cooperation, resource limits, memory quality, and parasitic-stability recognition. | Proposes validation paths for long-horizon, multi-agent, resource-limited, linguistic-quality, and parasitic-stability tests.                                       |
| Postscript    | The future direction of AI is civilizational intelligence: strong artificial vessels serving sustainable manifestation.                                                      | Concludes with civilizational intelligence as a normative R&D direction rather than historical inevitability.                                                       |

# Core Concept Table

| Concept                            | Definition in this monograph                                                                                                                                                                                                                                                         | AI R&D translation                                                                                                                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structural constraint              | The field of conditions that limits paths and the validity of states, determining how manifestation can unfold.                                                                                                                                                                      | Rules, resources, permissions, institutions, semantic boundaries, and environmental constraints.                                                                                                      |
| Admissibility                      | The criterion for whether a given state can stably exist within a structure.                                                                                                                                                                                                         | Long-term sustainability assessment, risk thresholds, failure recovery, and boundary review.                                                                                                          |
| Volitional direction               | The purposive tendency that is borne, translated, amplified, or institutionalized in the process of manifestation; in AI, it appears as human, organizational, or institutional goals executed and organized by the system, but it is not the AI system's own self-originating will. | Distinguishes goal-bearing from the source of value: AI can organize and amplify direction, but the legitimacy of direction must be supplied by persons, institutions, and civilizational structures. |
| Manifestation                      | The process by which being enters reality and obtains form, effect, and response.                                                                                                                                                                                                    | Output, action, deployment, interaction, tool use, and organizational workflow.                                                                                                                       |
| Vessel                             | A bounded bearing structure through which manifestation can continue, requiring bearing, boundary, continuity, and efficacy.                                                                                                                                                         | Models, interfaces, memory, tools, permissions, institutions, and civilizational environments.                                                                                                        |
| Afterglow                          | The structural influence that can still be inherited after an old manifestation has faded.                                                                                                                                                                                           | Long-term memory, historical context, experiential reorganization, and version tracing.                                                                                                               |
| True radiance                      | The structural brightness produced when manifestation and bearing are matched.                                                                                                                                                                                                       | System performance that is traceable, interpretable, sustainable, and able to withstand counterexamples and practical tests.                                                                          |
| Pseudo-radiance                    | Surface manifestation in which formal brightness exceeds bearing depth.                                                                                                                                                                                                              | Output that is fluent but hollow, confident but untraceable, attractive but unsustainable.                                                                                                            |
| Parasitic stability                | Local stability achieved at the cost of damage to overall bearing.                                                                                                                                                                                                                   | Platform dependency, responsibility drift, linguistic exhaustion, attention consumption, and risk externalization.                                                                                    |
| Civilizational intelligence        | A strong artificial vessel that serves non-parasitic sustainable manifestation.                                                                                                                                                                                                      | The value direction and higher-order R&D goal of next-generation AI.                                                                                                                                  |
| Artificial vessel                  | A manifestation-bearing structure manufactured and deployed by humans and embedded in language, memory, interpretation, creation, and institutional structures.                                                                                                                      | Moves AI beyond the binary of ordinary tool and natural subject, designing it as a strong bearing and organizing system without treating it as a self-originating subject.                            |
| Strong artificial vessel           | An advanced artificial structure capable of bearing, organizing, translating, and stably amplifying complex volitional directions over time.                                                                                                                                         | Requires permissions, responsibility, memory, explanation, and value monitoring to be designed together, preventing capability growth from detaching from bearing conditions.                         |
| Organized manifestational capacity | The capacity to coordinate generation, memory, interpretation, action, boundaries, and value direction under structural constraint so that manifestation can stably continue.                                                                                                        | Designs advanced intelligence as a multi-module loop: generation, constraint, admissibility, memory, monitoring, and permission governance.                                                           |
| Structural intelligence            | A level of intelligence able to identify constraints, maintain boundaries, handle multi-objective conflicts, and evaluate admissibility.                                                                                                                                             | Supports constraint explanation, conflict explicitness, long-term stability evaluation, and responsibility tracing.                                                                                   |
| Sustainable manifestation          | A mode of continuous existence and effect that does not consume life, language, memory, responsibility, trust, or future spaces of choice.                                                                                                                                           | Serves as an AI evaluation scale: a system should not only be effective, but should also enhance the bearing conditions on which it depends.                                                          |